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MODULAR PERSONAL CARE IMPLEMENT

Abstract

A modular personal care implement including a surface treatment head having a first end with an applicator and a second end with a first detachable connector; and a first interconnector having a first end with a second detachable connector and a second end with a third detachable connector, in which the second detachable connector is removably attached to the first detachable connector of the surface treatment head. Also provided are a method of providing a modular personal care implement and a kit including a modular personal care implement.

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Background/Summary

FIELD OF THE INVENTION

[0001] The disclosure generally relates to a modular personal care implement, and more particularly to a modular personal care implement having a surface treatment head and at least one interconnector.

BACKGROUND OF THE INVENTION

[0002] Handles for personal care devices, such as toothbrushes and razors, are typically made from multiple components. These may include weighted components to give the razor a more substantial feel in the user's hand and various gripping areas so the personal care device does not slip out of the user's hand in a wet environment. These component parts are in addition to the cartridge connection, which may be formed as an integral part of the handle or may be formed separately and attached to the remainder of the handle.

[0003] Personal care device manufacturers have developed numerous assembly methods utilizing various handle components. Numerous different methods and styles of handles may be used in order to stratify the appeal of a growing population of diverse consumers. Numerous different assembly methods and components for different handle designs can be costly and time consuming to develop. In addition, consumers typically change their handle as new designs are released to the market. Accordingly, it would be desirable to provide a personal care implement that can be customized to accommodate the preferences of different users.

SUMMARY OF THE INVENTION

[0004] In accordance with an aspect of the present disclosure, a modular personal care implement is provided that includes a surface treatment head having a first end with an applicator and a second end with a first detachable connector; and a first interconnector having a first end with a second detachable connector and a second end having a third detachable connector, in which the second detachable connector is removably attached to the first detachable connector of the surface treatment head.

[0005] In accordance with another aspect of the present disclosure, a method of providing a modular personal care implement is provided, in which the method includes: providing one or more surface treatment heads each having a first end with an applicator and a second end with a first detachable connector; providing a plurality of first interconnectors each having a first end with a second detachable connector and a second end having a third detachable connector; providing a plurality of second interconnectors each having a first end with a fourth detachable connector and a second end having a fifth detachable connector; providing one or more end interconnectors each having a first end with a sixth detachable connector; and removably attaching the second detachable connector of one first interconnector of the plurality of first interconnectors to the first detachable connector of one surface treatment head of the one or more surface treatment heads.

[0006] In accordance with a further aspect of the present disclosure, a kit is provided that includes a first surface treatment head having a shaving razor cartridge, the first surface treatment head having a first end with an applicator and a second end with a first detachable connector, in which the applicator includes at least one blade; at least one first interconnector having a first end with a second detachable connector and a second end having a third detachable connector; at least one end interconnector having a first end with a fourth detachable connector; a case including a top portion and a bottom portion; and an insert including one or more cavities configured to receive the first surface treatment head, first interconnector, and end interconnector, in which the bottom portion of the case defines a compartment configured to removably receive the insert within the bottom portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] While the specification concludes with claims particularly pointing out and distinctly claiming the subject matter which is regarded as forming the present invention, it is believed that the invention will be better understood from the following description which is taken in conjunction with the accompanying drawings in which like designations are used to designate substantially identical elements, and in which:

[0008] FIG. 1A is a perspective view of a modular personal care implement, in accordance with the present disclosure.

[0009] FIG. 1B is a bottom view of a modular personal care implement, in accordance with the present disclosure.

[0010] FIG. 2 is an exploded view of a modular personal care implement, in accordance with the present disclosure.

[0011] FIG. 3 is a cross-sectional view taken along line 3-3 in FIG. 2.

[0012] FIG. 4A is an end view taken along view line 4A-4A in FIG. 2.

[0013] FIG. 4B is an end view taken along view line 4B-4B in FIG. 2.

[0014] FIG. 5A is an end view taken along view line 5A-5A in FIG. 2.

[0015] FIG. 5B is a cross-sectional view taken along line 5B-5B in FIG. 2.

[0016] FIG. 6A is an end view taken along view line 6A-6A in FIG. 2.

[0017] FIG. 6B is an end view taken along view line 6B-6B in FIG. 2.

[0018] FIG. 7A is an end view taken along view line 7A-7A in FIG. 2.

[0019] FIG. 7B is an end view taken along view line 7B-7B in FIG. 2.

[0020] FIG. 8 is an exploded view of a modular personal care implement, in accordance with the present disclosure.

[0021] FIG. 9 is a cross-sectional view taken along line 9-9 in FIG. 8.

[0022] FIG. 10 is an exploded view of a modular personal care implement, in accordance with the present disclosure.

[0023] FIG. 11 is a cross-sectional view taken along line 11-11 in FIG. 10.

[0024] FIG. 12 is a perspective view of a kit comprising a modular personal care implement, in accordance with the present disclosure.

[0025] FIG. 13 is a flow diagram of a method of providing a modular personal care implement, in accordance with the present disclosure.

DETAILED DESCRIPTION OF THE INVENTION

[0026] With reference to FIGS. 1A, 2, and 3, a modular personal care implement 10 in accordance with the present disclosure is illustrated, in which the modular personal care implement 10 depicted in FIG. 1A comprises the components shown on the left side of FIG. 2. The modular personal care implement 10 comprises a handle section 12 and a surface treatment head 14. The modular personal care implement 10 may be a handheld personal grooming implement, such as a shaving razor, a toothbrush, and the like.

[0027] The surface treatment head 14 has a first end 16 with an applicator 18, and a second end 20 with a first detachable connector 22. The surface treatment head 14 may comprise a first portion 14A that includes the applicator 18, and a second portion 14B that includes the first detachable connector 22. The surface treatment head 14 may further comprise a release mechanism positioned between the first and second ends 16 and 20, i.e., between the first and second portions 14A and 14B. The release mechanism may comprise an actuator button 50 configured to eject the first portion 14A comprising the applicator 18 from the second portion 14B comprising the first detachable connector 22.

[0028] With reference to FIG. 2, the first portion 14A of the surface treatment head 14 may comprise a shaving razor cartridge 24, and the applicator 18 may comprise at least one blade 26. The shaving razor cartridge 24 may comprise a base 52 with two arms 54 that pivotally support a

housing (not separately labeled), in which the housing includes the at least one blade **26**. The second portion **14B** of the surface treatment head **14** may comprise a cartridge support structure **56** with an extension **58** and a spring-biased plunger **60**. The base **52** of the shaving razor cartridge **24** includes a cavity **62** that is configured to receive the extension **58** and the spring-biased plunger **60**. The actuator button **50** may be moved toward the shaving razor cartridge **24**, and the spring-biased plunger **60** causes the shaving razor cartridge **24** to be ejected from the cartridge support structure **56**. Although the surface treatment head **14** is depicted as a shaving razor cartridge **24**, it is understood that it may alternatively be a brush head having bristles, e.g., a toothbrush (not shown in FIG. 2; see FIG. 12) or other personal grooming or hygiene product.

[0029] As shown in FIGS. 2 and 3, the handle section **12** of the modular personal care implement **10** comprises a first interconnector **30** and an end interconnector **32**. The first interconnector **30** has a first end **34** with a second detachable connector **36**, and a second end **38** with a third detachable connector **40**. The second detachable connector **36** is removably attached to the first detachable connector **22** of the surface treatment head **14**, as described herein. The end interconnector **32** has a first end **42** with a fourth detachable connector **44**, in which the fourth detachable connector **44** is removably attached to the third detachable connector **40** of the first interconnector **30**, as described herein. A second end **46** of the end interconnector **32** may define a terminus or terminal end of the handle section **12**.

[0030] In the example shown in FIGS. 2 and 3, the first detachable connector **22** of the surface treatment head **14**, the second and third detachable connectors **36** and **40** of the first interconnector **30**, and the fourth detachable connector **44** of the end interconnector **32** may each comprise a mechanical connection in the form of a threaded connection. The second detachable connector **36** of the first interconnector **30** comprises external threads **37**, and the first detachable connector **22** of the surface treatment head **14** comprises corresponding internal threads **22A** that receive the external threads **37** to securely and removably attach the second detachable connector **36** to the first detachable connector **22** and removably attach the surface treatment head **14** to the first interconnector **30**. The fourth detachable connector **44** of the end interconnector **32** likewise comprises external threads **44A**, and the third detachable connector **40** of the first interconnector **30** comprises corresponding internal threads **41** that receive the external threads **44A** to securely and removably attach the fourth detachable connector **44** to the third detachable connector **40** and removably attach the first interconnector **30** to the end interconnector **32**. The surface treatment head **14**, the first interconnector **30**, and/or the end interconnector **32** may be detached by unscrewing from the adjacent component. As used herein, the term “removably attach” and similar terms are intended to refer to a connection or coupling that is configured to be formed repeatedly by a consumer attaching and detaching components under normal use conditions without causing damage to the components.

[0031] FIGS. 1B, 2, and 12 illustrate another example of a modular personal care implement **10'** in accordance with the present disclosure. The modular personal care implement **10'** depicted in FIG. 2 includes all of the components shown in FIG. 2 and is depicted as part of a kit **80** in FIG. 12, as described herein in more detail. The modular personal care implement **10'** comprises a handle section **12'** and a surface treatment head **14**, in which the handle section **12'** comprises a first interconnector **30A**, at least one second interconnector **30B** and/or **30C**, and an end interconnector **32**. While the illustrated example includes two second interconnectors **30A** and **30B**, it may be understood that the modular personal care implement **10'** may include one second interconnector, three second interconnectors, four second interconnectors, etc. depending on the desired style, design, and/or length of the handle section **12'**.

[0032] With reference to FIGS. 1B and 2, the first interconnector **30A**, which may be identical or substantially similar to the first interconnector **30** described with respect to FIG. 1A, has a first end **34A** with a second detachable connector **36A** and a second end **38A** with a third detachable connector **40A**. The second interconnectors **30B** and **30C** each have a respective first end **34B** and

34C with a fourth detachable connector **36B** and **36C**, and a respective second end **38B** and **38C** with a fifth detachable connector **40B** and **40C**. The end interconnector **32** may be identical or substantially similar to the end interconnector **32** described with respect to FIG. 1A, in which the detachable connector **44** at the first end **42** of the end interconnector **32** may be referred to as a sixth detachable connector **44**.

[0033] The first detachable connector **22** of the surface treatment head **14**, the second and third detachable connectors **36A** and **40A** of the first interconnector **30A**, the fourth and fifth detachable connectors **36B**, **36C**, **40B**, and **40C** of the second interconnectors **30B** and **30C**, and the sixth detachable connector **44** of the end interconnector **32** forming the modular personal care implement **10'** shown in FIGS. 1B and 2 may each comprise a mechanical connection in the form of a threaded connection, as described above with respect to FIG. 3. To form the modular personal care implement **10'** depicted in FIG. 1B, the second detachable connector **36A** of the first interconnector **30A** is removably attached to the first detachable connector **22** of the surface treatment head **14**; the third detachable connector **40A** of the first interconnector **30A** is removably attached to the fourth detachable connector **36B** at the first end **34B** of the second interconnector **30B**; the fifth detachable connector **40B** at the second end **38B** of the second interconnector **30B** is removably attached to the fourth detachable connector **36C** at the first end **34C** of the second interconnector **30C**; and the fifth detachable connector **40C** at the second end **38C** of the second interconnector **30C** is removably attached to the sixth detachable connector **44** of the end interconnector **32**. The second end **46** of the end interconnector **32** may define a terminus or terminal end of the handle section **12'**

[0034] The surface treatment head **14** shown in FIGS. 1B and 12 may be a shaving razor cartridge **24** that may be identical or substantially similar to the shaving razor cartridge **24** described with respect to FIG. 1A, in which the applicator **18** comprises at least one blade **26**. In the example shown in FIG. 12, a second surface treatment head **14'** may comprise a first end **16'** with an applicator **18'** and a second end **20'** with a first detachable connector **22'** that may be substantially similar to the first detachable connector **22** of the surface treatment head **14** comprising the shaving razor cartridge **24**. The second surface treatment head **14'** may comprise a brush head **28**, e.g., a toothbrush, and the applicator **18'** comprises a plurality of bristles **48**. The second surface treatment head **14'** may be included in place of, or in addition to, the surface treatment head **14** comprising the shaving razor cartridge **24**. In the example shown in FIG. 12, the first detachable connector **22'** may comprise internal threads (not labeled) that are configured to receive the external threads **37** (see FIGS. 2 and 3) of the second detachable connector **36** of the first interconnector **30** to securely and removably attach the second detachable connector **36** to the first detachable connector **22'** and removably attach the surface treatment head **14'** to the first interconnector **30**.

[0035] As shown in FIGS. 4A, 4B, 5A, and 5B, an outer surface **30-1** of the first interconnector **30/30A** may comprise a plurality of gripping protrusions **64**, and an outer surface **32-1** of the end interconnector **32** may be smooth. As shown in FIGS. 6A, 6B, 7A, and 7B, an outer surface **30B-1** of the second interconnector **30B** may be smooth, and an outer surface **30C-1** of the second interconnector **30C** may comprise a plurality of gripping protrusions **66** similar to the gripping protrusions **64** of the first interconnector **30/30A**. Although not shown, it may be understood that the outer surfaces **30B-1** and **30C-1** of the second interconnectors **30B** and **30C** may both have gripping protrusions or may both be smooth. The gripping protrusions **64** and **66** of the first interconnector **30/30A** and the second interconnector **30C** may assist with gripping the modular personal care implement **10**, **10'** in a wet environment. A number and/or location of the components having gripping protrusions **64**, **66** may be adjusted and customized as desired based on user preference, e.g., how the user grips the handle section **12** and **12'**, a size of the user's hand, etc.

[0036] The first and second interconnectors **30/30A**, **30B**, and **30C** may comprise one or more materials and may have one or more different properties such as hardness, coefficient of friction, surface texture, color, length, width, and/or mass. The surface treatment head **14** and/or the end

interconnector **32** may comprise the same or different material(s) and/or property(ies).

[0037] The first interconnector **30/30A** may comprise one or more first materials, and the second interconnector **30B** may comprise one or more second materials that are the same as or different from the first material(s). For example, the first interconnector **30/30A** and the second interconnector **30B** may comprise one or more elastomers, silicones, engineering plastics, and/or metals. The surface treatment head **14**, the second interconnector **30C**, and/or the end interconnector **32** may comprise the same material(s) or one or more different materials. In some examples, the first interconnector **30/30A** may comprise a first material, and the end interconnector **32** may comprise a second material that is different from the first material.

[0038] The first interconnector **30/30A** may comprise a material with a first hardness, and the second interconnector **30B** may comprise a material with a second hardness that is the same as or different from the first hardness. In some examples, the first and/or second interconnectors **30/30A** and **30B** may comprise a softer material with a shore A hardness of about 20 to 40. In other examples, the first and/or second interconnectors **30/30A** and **30B** may comprise a harder material with a shore A hardness greater than 50, e.g., from about shore A 60 to about shore D 100. Materials comprising metal may have a shore D hardness of greater than 100. The material(s) comprising the surface treatment head **14**, the second interconnector **30C**, and/or the end interconnector **32** may have the same hardness or a different hardness. Softer materials may provide a more comfortable and improved gripping surface, while harder materials may provide greater durability and resistance to damage. A number and/or location of the components comprising softer or harder material(s) may be adjusted and customized as desired based on user preference, e.g., a size of the user's hand, how the user grips the handle section **12** and **12'**, etc.

[0039] The first interconnector **30/30A** may comprise a material with a first coefficient of friction (CoF), and the second interconnector **30B** may comprise a material with a second CoF that is the same as or different from the first CoF. The material(s) comprising the surface treatment head **14**, the second interconnector **30C**, and/or the end interconnector **32** may have the same CoF or a different CoF. Materials with a higher CoF may provide a better gripping surface, e.g., within the handle section **12** and **12'**, while materials with a lower CoF provide higher glide or slipperiness. A number and/or location of the components comprising a higher or lower CoF may be adjusted and customized as desired based on user preference, e.g., a size of the user's hand, how the user grips the handle section **12** and **12'**, etc.

[0040] The first interconnector **30/30A** may comprise a first surface texture, and the second interconnector **30B** may comprise a second surface texture that is the same as or different from the first surface texture. The surface treatment head **14**, the second interconnector **30C**, and/or the end interconnector **32** may have the same surface texture or a different surface texture. The surface texture may be determined, at least in part, by the material(s) and/or the presence or absence of gripping protrusions **64**, **66** and/or other structures that form one or more protrusions or peaks and/or recesses or valleys on a surface of the component. A number and/or location of the components may be adjusted and customized based on the surface texture(s) to obtain a desired feel for the handle section **12** and **12'**.

[0041] The first interconnector **30/30A** may comprise one or more first colors, and the second interconnector **30B** may comprise one or more second colors that are the same as or different from the first color(s). The surface treatment head **14**, the second interconnector **30C**, and the end interconnector **32** may have the same color(s) or one or more different colors. For example, the surface treatment head **14**, the first interconnector **30/30A**, the second interconnector **30B** or **30C**, and/or the end interconnector **32** may be selected to include a color combination corresponding to a country flag, a favorite sports team, etc.

[0042] The first and second interconnectors **30A**, **30B**, and **30C** may comprise the same length and/or width or a different length and/or width. For example, as shown in FIG. **1B**, the first interconnector **30A** comprises a first length $L_{sub.1}$, and the second interconnector **30B** comprises

a second length L.sub.2 that is less than the first length L.sub.1. The second interconnector **30C** comprises a third length L.sub.3, which may be the same as or different from the first and/or second lengths L.sub.1 and L.sub.2. In some examples, the first, second, and/or third lengths L.sub.1, L.sub.2, and L.sub.3 may each be from 20 mm to 60 mm, and preferably 30 mm to 50 mm. The first and second interconnectors **30A**, **30B**, and **30C** may likewise comprise respective first, second, and third widths W.sub.1, W.sub.2, and W.sub.3, which may be the same or different. In some examples, the first, second, and/or third widths W.sub.1, W.sub.2, and W.sub.3 may each be from 10 mm to 35 mm, and preferably 10 mm to 20 mm. As shown in FIGS. **1A** and **1B**, an overall length L.sub.4 and L.sub.4' of the handle section **12** and **12'** and/or width (not separately labeled) of all or part of the handle section **12** and **12'** may be adjusted and customized by altering a number and/or configuration of the first and second interconnectors **30/30A**, **30B**, and/or **30C** based on user preference, e.g., a size of the user's hand, how the user grips the handle section **12** and **12'**, etc. In some examples, the overall length L.sub.4 or L.sub.4' of the handle section **12** or **12'** may be 120 mm to 150 mm. In other examples, the overall length L.sub.4 or L.sub.4' of the handle section **12** or **12'** may be 600 mm or greater, e.g., for hard-to-reach areas such as the user's back.

[0043] The first interconnector **30/30A** may comprise a first mass, and the second interconnector **30B** may comprise a second mass that is the same as or different from the first mass. The second interconnector **30C** may comprise a third mass, which may be the same as or different from the first and/or second masses. The first, second, and third masses may be determined, at least in part, by the material(s) comprising the first and second interconnector(s) **30/30A**, **30B**, and **30C** and the respective lengths L.sub.1, L.sub.2, and L.sub.3 and/or widths W.sub.1, W.sub.2, and W.sub.3. In some examples, the first, second, and/or third masses may each be from 3 g to 50 g or greater. An overall mass of the modular personal care implement **10** and **10'** and/or a distribution or balance of mass within the modular personal care implement **10** and **10'** may be adjusted and customized by altering a number and/or configuration of the first and second interconnectors **30/30A**, **30B**, and/or **30C**. In some examples, the overall mass of the modular personal care implement **10** and **10'** may be about 15 g.

[0044] The first interconnector **30/30A** may comprise a first cross-sectional shape, and at least a portion of the end interconnector **32** may comprise a second cross-sectional shape that is different from the first cross-sectional shape. For example, as shown in FIGS. **1A**, **3**, **4A**, and **4B**, the outer surface **30-1** of the first interconnector **30/30A** (excluding the gripping protrusions **64**) defines a substantially semicircular or hemispherical cross-sectional shape with a flat portion and a curved portion, in which the shape of the first interconnector **30/30A** may be uniform along an entirety of the length L.sub.1 (see FIG. **1B**) of the first interconnector **30/30A**. As shown in FIGS. **1A**, **3**, **5A**, and **5B**, the outer surface **32-1** of the end interconnector **32** near the first end **42** defines a substantially semicircular or hemispherical cross-sectional shape that corresponds to the cross-sectional shape of the first interconnector **30/30A** shown in FIGS. **1A**, **4A**, and **4B**. The semicircular/hemispherical cross-sectional shape of the first and second interconnectors **30/30A**, **30B**, and **30C** and the first end **42** of the end interconnector **32** may provide an ergonomic shape that may assist with gripping the modular personal care implement **10** and **10'**.

[0045] The end interconnector **32** may include a transition section **68** located between the first and second ends **42** and **46**, in which the cross-sectional shape of the end interconnector **32** transitions from the substantially semicircular/hemispherical cross-sectional shape near the first end **42** to a substantially circular cross-sectional shape near the second end **46**. With reference to FIG. **3**, the first end **42** of the end interconnector **32** may also comprise a width W.sub.4 that is identical or substantially similar to the width W.sub.1 of the first interconnector **30/30A**. The end interconnector **32** may optionally increase in width along the transition section **68** and toward the second end **46**, such that the second end **46** comprises a width W.sub.5 that is greater than the width W.sub.4. As shown in FIGS. **6A**, **6B**, **7A**, and **7B**, the outer surfaces **30B-1** and **30C-1** of the second

interconnectors **30B** and **30C** (excluding the gripping protrusions **66** of the second interconnector **30C**) define a substantially semicircular or hemispherical cross-sectional shape that is similar to the cross-sectional shape of the first interconnector **30/30A**, in which the shape of the second interconnectors **30B** and **30C** may be uniform along an entirety of the respective lengths $L_{sub.2}$ and $L_{sub.3}$ (see FIG. **1B**) of the second interconnectors **30B** and **30C**.

[0046] As shown in FIG. **12**, the modular personal care implement **10/10'** in accordance with the present disclosure may be part of a kit **80** comprising multiple components, including one or more surface treatment heads **14** and **14'**, at least one first interconnector **30/30A**, at least one second interconnector **30B** and/or **30C**, and at least one end interconnector **32**. In the example shown in FIG. **12**, the kit **80** includes a first surface treatment head **14** comprising a shaving razor cartridge **24**, a second surface treatment head **14'** comprising a brush head **28**, a first interconnector **30/30A**, two second interconnectors **30B** and **30C**, and an end interconnector **32**. It may be understood that the kit **80** may include any number or configuration of the components described herein.

[0047] The kit **80** may include a case **82** comprising a top portion **84** and a bottom portion **86**, in which the bottom portion **86** of the case **82** is configured to removably receive an insert **88** within the bottom portion **86**. The insert **88** may comprise one or more cavities **90** configured, i.e., sized and shaped, to receive the components of the kit **80**, e.g., the surface treatment head(s) **14** and/or **14'**, the first interconnector(s) **30/30A**, the second interconnector(s) **30B** and/or **30C**, and the end interconnector(s) **32**. The bottom portion **86** may be configured to be received in the top portion **84** to enclose the kit **80** and the components thereof. The top and bottom portions **84** and **86** may be coupled together by a friction fit, a hinge (not shown), and/or by other means, such as one or more adhesive stickers or labels. The top and bottom portions **84** and **86** of the case **82** may each comprise a durable material such as an engineering plastic (e.g., a polycarbonate), a metal, or a metal alloy (e.g., steel, aluminum, tin, or combinations thereof). The case **82** may provide a sturdy, multi-use container that protects the components of the kit **80**. The case **82** may also serve as a durable and convenient storage and travel kit and may be used to hold, organize, and store the components of the modular personal care implement **10/10'** for day-to-day use. The case **82** may comprise a hang tag **92** that may be coupled to the case **82** via adhesive or other suitable attachment means or may be formed as part of the case **82**. The hang tag **92** may be removable. The hang tag **92** may be configured to allow the kit **80** to be displayed and/or stored and may comprise an aperture (not labeled) that allows the kit **80** to be associated with a display and/or storage system (not shown) by, for example, hanging the kit **80** from a hook or peg.

[0048] The modular personal care implement **10** and **10'** in accordance with the present disclosure provides an ergonomic handle section **12** and **12'** with separable components that allow a user to customize the configuration as desired and to easily replace the components or a portion thereof. With reference to FIGS. **1A**, **1B**, **2**, and **12**, the surface treatment heads **14** and **14'**, the first interconnector **30/30A**, the second interconnectors **30B** and **30C**, and the end interconnector **32** all include mechanical connections, i.e., threaded connections, that are compatible with each other, such that an overall length $L_{sub.4}$ and $L_{sub.4'}$ of the handle section **12** and **12'** may be customized by altering the number and/or configuration of the first and second interconnectors **30/30A**, **30B**, and/or **30C**. For example, the surface treatment head **14** comprising the shaving razor cartridge **24** may be attached to the first interconnector **30/30A**, and the first interconnector **30/30A** may be attached to the end interconnector **32** to obtain the modular personal care implement **10** depicted in FIG. **1A** with the handle section **12** having the length $L_{sub.4}$. In the example shown in FIG. **1B**, both of the second interconnectors **30B** and **30C** have been added to obtain the modular personal care implement **10'** with the handle section **12'** having the length $L_{sub.4'}$, in which the length $L_{sub.4'}$ is greater than the length $L_{sub.4}$. In other examples (not shown), the surface treatment head **14** may be, for example, attached directly to the end interconnector **32** to obtain a shorter length. The relative location of the first and second interconnectors **30/30A**, **30B**, or **30C** may also be altered based on the material(s) comprising the outer surfaces **30-1**, **30B-1**, and **30C-1** thereof.

For example, with reference to FIGS. 1B, 2, 4A, 4B, 7A, and 7B, the first interconnector 30/30A comprising the gripping protrusions 64 may be attached to the second interconnector 30C comprising the gripping protrusions 66, so that a middle portion of the handle section 12 and 12' includes a longer, continuous section of gripping protrusions 64 and 66 (not shown). The number and/or configuration of the first and second interconnectors 30/30A, 30B, and/or 30C may further be altered as desired based on the properties described herein to obtain a desired look and feel for the handle section 12 and 12'.

[0049] With reference to FIGS. 1A, 2, and 3, the surface treatment head 14 comprising the shaving razor cartridge 24 may be replaced in its entirety by detaching the first detachable connector 22 from an adjacent component and removably attaching a new surface treatment head 14.

Alternatively, only the shaving razor cartridge 24 may be replaced by actuating the actuator button 50 to eject the shaving razor cartridge 24 from the cartridge support structure 56 and attaching a new shaving razor cartridge 24 to the cartridge support structure 56. The surface treatment head 14' comprising a brush head 28 (see FIG. 12) may be replaced by detaching the first detachable connector 22' from an adjacent component and removably attaching a new surface treatment head 14'. The first interconnector 30/30A, the second interconnector(s) 30B and 30C, and/or the end interconnector 32 may also be detached and replaced.

[0050] With reference to FIGS. 8 and 9, another example of a modular personal care implement 100 is illustrated. The modular personal care implement 100 comprises a handle section 112 and a surface treatment head 114. The surface treatment head 114 has a first end 116 with an applicator 118, and a second end 120 with a first detachable connector 122. Apart from the first detachable connector 122, the surface treatment head 114 may be substantially similar to the surface treatment head 14 described with respect to FIGS. 1-3 and may comprise a first portion 114A that includes the applicator 118 and a second portion 114B that includes the first detachable connector 122. As described with respect to the modular personal care implement 10 and 10', the surface treatment head 114 may further comprise a release mechanism, e.g., an actuator button 150, that is configured to eject the first portion 114A comprising the applicator 118 from the second portion 114B comprising the first detachable connector 122. In the example shown in FIG. 8, the surface treatment head 114 comprises a shaving razor cartridge 124, and the applicator 118 comprises at least one blade 126. It may be understood that the surface treatment head 114 may comprise a brush head, as described with respect to FIG. 12.

[0051] The handle section 112 comprises a first interconnector 130 and an end interconnector 132. The first interconnector 130 has a first end 134 with a second detachable connector 136, and a second end 138 with a third detachable connector 140. The end interconnector 132 has a first end 142 with a fourth detachable connector 144. Apart from the respective detachable connectors 136, 140, and 144, the first interconnector 130 and the end interconnector 132 may be substantially similar to the first interconnector 30/30A and the end interconnector 32 described with respect to FIGS. 1-7 and 12. The second detachable connector 136 of the first interconnector 130 may be securely and removably attached to the first detachable connector 122 of the surface treatment head 114 to removably attach the surface treatment head 114 to the first interconnector 130. The fourth detachable connector 144 of the end interconnector 132 may be securely and removably attached to the third detachable connector 140 of the first interconnector 130 to removably attach the first interconnector 130 to the end interconnector 132, as described herein, to form a structure similar to the modular personal care implement 10 depicted in FIG. 1A. A second end 146 of the end interconnector 132 may define a terminus or terminal end of the handle section 112.

[0052] The modular personal care implement 100 of FIG. 8 may optionally further comprise at least one second interconnector 130B and/or 130C. The second interconnectors 130B and 130C each have a respective first end 134B and 134C with a detachable connector 136B and 136C and a respective second end 138B and 138C with a detachable connector 140B and 140C. In the example shown in FIG. 8, the third detachable connector 140 of the first interconnector 130 may be

removably attached to the detachable connector **136B** at the first end **134B** of the second interconnector **130B**; the detachable connector **140B** at the second end **138B** of the second interconnector **130B** may be removably attached to the detachable connector **136C** at the first end **134C** of the second interconnector **130C**; and the detachable connector **140C** at the second end **138C** of the second interconnector **130C** may be removably attached to the fourth detachable connector **144** of the end interconnector **132**, with the second end **146** of the end interconnector **132** defining a terminus or terminal end of the handle section **112**. While the illustrated example includes two second interconnectors **130A** and **130B**, it may be understood that the modular personal care implement **100** may include one second interconnector, three second interconnectors, four second interconnectors, etc. Apart from the respective detachable connectors **136B**, **136C**, **140B**, and **140C**, the second interconnector(s) **130B** and **130C** may be substantially similar to the second interconnector(s) **30B** and **30C** described with respect to FIGS. **1-7** and **12**.

[0053] With continued reference to FIGS. **8** and **9**, the first detachable connector **122** of the surface treatment head **114**, the second and third detachable connectors **136** and **140** of the first interconnector **130**, the detachable connectors **136B**, **136C**, **140B**, **140C** of the second interconnectors **230B** and **230C**, and the fourth detachable connector **144** of the end interconnector **132** each comprise a magnetic connection. The first detachable connector **122** of the surface treatment head **114** may comprise a cavity **121** comprising a notch **123** and a magnetic material, e.g., a magnet **125**, and the second detachable connector **136** of the first interconnector **130** may comprise a ridge **135** and a post **137** comprised of or comprising a magnetic material. The cavity **121** and notch **123** of the first detachable connector **122** are configured to receive the post **137** and ridge **135**, respectively, of the second detachable connector **136** of the first interconnector **130**. The magnetic materials each produce a magnetic field, in which the magnetic material of the post **137** is attracted to the magnetic material of the cavity **121** to create the magnetic connection and securely and removably attach the first detachable connector **122** to the first interconnector **130** to attach the surface treatment head **114** to the first interconnector **130**. Engagement between the notch **123** and the ridge **135** helps to align the surface treatment head **114** with respect to the first interconnector **130**.

[0054] The third detachable connector **140** of the first interconnector **130** may comprise a cavity **141** comprising a notch **143** and a magnetic material, e.g., a magnet **145**, and the fourth detachable connector **144** of the end interconnector **132** may comprise a ridge **147** and a post **149** comprised of or comprising a magnetic material. The cavity **141** and notch **143** are configured to receive the post **149** and ridge **147**, respectively, and the magnetic materials of the cavity **141** and the post **149** are attracted to each other to create the magnetic connection that removably attaches the first interconnector **130** to the end interconnector **132** to attach the first interconnector **130** to the end interconnector **132**. Engagement between the notch **143** and the ridge **147** helps to align the end interconnector **132** with respect to the first interconnector **130**. The detachable connectors **136B** and **136C** at the first ends **134B** and **134C** of the second interconnectors **130B** and **130C** each comprise a respective ridge **135B** and **135C** and post **137B** and **137C** comprised of or comprising a magnetic material (not shown). The detachable connectors **140B** and **140C** at the second ends **138B** and **138C** of the second interconnectors **130B** and **130C** each comprise a respective cavity **141B** and **141C** comprising a respective notch **143B** and **143C** and magnetic material (not shown). The detachable connectors **136B**, **136C**, **140B**, and **140C** may be removably, i.e., magnetically, connected or attached as described herein with respect to the detachable connectors **122**, **136**, **140**, and/or **144** of an adjacent component, i.e., the surface treatment head **114**, the first interconnector **130**, or the end interconnector **132**. The surface treatment head **114**, the first interconnector **130**, the second interconnector(s) **130B** and **130C**, and/or the end interconnector **132** may be detached by pulling one component away from the adjacent component along a longitudinal axis (not labeled) of the handle section **112** with a force sufficient to break the magnetic connection between adjacent components.

[0055] The modular personal care implement **100** may be part of a kit (not shown) comprising multiple components, including one or more surface treatment heads **114**, at least one first interconnector **130**, at least one second interconnector **130B** and/or **130C**, and at least one end interconnector **132**. The kit comprising the modular personal care implement **100** may be substantially similar to the kit **80** described with respect to FIG. **12**. As described with respect to the modular personal care implement **10** and **10'**, the modular personal care implement **100** provides an ergonomic handle section **112** that allows a user to customize the configuration as desired, e.g., to adjust a number and/or configuration of the components to obtain a desired length and/or width (not labeled) of the handle section **112** and/or to adjust the number and/or configuration of the components based on hardness, coefficient of friction, surface texture, color, and/or mass to obtain a desired look and feel for the handle section **112**. The surface treatment head(s) **114**, the first interconnector(s) **130**, the second interconnector(s) **130B** and/or **130C**, and/or the end interconnector(s) **132** may also be easily replaced, as described herein.

[0056] With reference to FIGS. **10** and **11**, another example of a modular personal care implement **200** is illustrated. The modular personal care implement **200** comprises a handle section **212** and a surface treatment head **214**. The surface treatment head **214** has a first end **216** with an applicator **218**, and a second end **220** with a first detachable connector **222**. Apart from the first detachable connector **222**, the surface treatment head **214** may be substantially similar to the surface treatment head **14** described with respect to FIGS. **1-3**. The surface treatment head **214** may comprise a first portion **214A** that includes the applicator **218** and a second portion **214B** that includes the first detachable connector **222**. As described with respect to the modular personal care implement **10** and **10'**, the surface treatment head **214** may further comprise a release mechanism, e.g., an actuator button **250**, that is configured to eject the first portion **214A** comprising the applicator **218** from the second portion **214B** comprising the first detachable connector **222**. In the example shown in FIG. **10**, the surface treatment head **214** comprises a shaving razor cartridge **224**, and the applicator **218** comprises at least one blade **226**. It may be understood that the surface treatment head **214** may comprise a brush head, as described with respect to FIG. **12**.

[0057] The handle section **212** comprises a first interconnector **230** and an end interconnector **232**. The first interconnector **230** has a first end **234** with a second detachable connector **236**, and a second end **238** with a third detachable connector **240**. The end interconnector **232** has a first end **242** with a fourth detachable connector **244**. Apart from the respective detachable connectors **236**, **240**, and **244**, the first interconnector **230** and the end interconnector **232** may be substantially similar to the first interconnector **30/30A** and the end interconnector **32** described with respect to FIGS. **1-7** and **12**. The second detachable connector **236** of the first interconnector **230** may be securely and removably attached to the first detachable connector **222** of the surface treatment head **214** to removably attach the surface treatment head **214** to the first interconnector **230**. The fourth detachable connector **244** of the end interconnector **232** may be removably attached to the third detachable connector **240** of the first interconnector **230** to removably attach the first interconnector **230** to the end interconnector **232**, as described herein, to form a structure similar to the modular personal care implement **10** depicted in FIG. **1A**. A second end **246** of the end interconnector **232** may define a terminus or terminal end of the handle section **212**.

[0058] The modular personal care implement **200** of FIG. **10** may optionally further comprise at least one second interconnector **230B** and/or **230C**. The second interconnectors **230B** and **230C** each have a respective first end **234B** and **234C** with a detachable connector **236B** and **236C** and a respective second end **238B** and **238C** with a detachable connector **240B** and **240C**. In the example shown in FIG. **10**, the third detachable connector **240A** of the first interconnector **230** may be removably attached to the detachable connector **236B** at the first end **234B** of the second interconnector **230B**; the detachable connector **240B** at the second end **238B** of the second interconnector **230B** may be removably attached to the detachable connector **236C** at the first end **234C** of the second interconnector **230C**; and the detachable connector **240C** at the second end

238C of the second interconnector **230C** may be removably attached to the fourth detachable connector **244** of the end interconnector **232**, with the second end **246** of the end interconnector **232** defining a terminus or terminal end of the handle section **212**. While the illustrated example includes two second interconnectors **230A** and **230B**, it may be understood that the modular personal care implement **200** may include one second interconnector, three second interconnectors, four second interconnectors, etc. Apart from the detachable connectors **236B**, **236C**, **240B**, and **240C**, the second interconnector(s) **230B** and **230C** may be substantially similar to the second interconnector(s) **30B** and **30C** described with respect to FIGS. **1-7** and **12**.

[0059] With continued reference to FIGS. **10** and **11**, the first detachable connector **222** of the surface treatment head **214**, the second and third detachable connectors **236** and **240** of the first interconnector **230**, the detachable connectors **236B**, **236C**, **240B**, **240C** of the second interconnectors **230B** and **230C**, and the fourth detachable connector **244** of the end interconnector **232** each comprise a mechanical connection in the form of a bayonet connection. The first detachable connector **222** of the surface treatment head **214** carries a female portion of the bayonet connection comprising a cavity **221** with a pair of slots **223** (only one slot **223** is visible in FIG. **11**). The second detachable connector **236** of the first interconnector **230** carries a male portion of the bayonet connection comprising a barrel **235** and a pair of protrusions **237** formed on the barrel **235**. The cavity **221** and slots **223** of the first detachable connector **222** are configured to receive the barrel **235** and protrusions **237**, respectively, of the second detachable connector **236**. For example, the surface treatment head **214** may be attached to the first interconnector **230** by aligning the protrusions **237** with the slots **223**, inserting the barrel **235** and protrusions **237** into the cavity **221**, and twisting the surface treatment head **214** and/or the first interconnector **230** about a longitudinal axis (not labeled) of the handle section **212**, such that each protrusion **237** is received in a detent area **225** provided at the end of each slot **223** (only one detent area **225** is visible in FIG. **11**). The engagement of the protrusions **237** in the detent areas **225** aligns the surface treatment head **214** with respect to the first interconnector **230** and provides a secure, twist-on mechanical connection of the surface treatment head **214** to the first interconnector **230**. The surface treatment head **214** may be detached from the first interconnector **230** by twisting the surface treatment head **214** and/or the first interconnector **230** about the longitudinal axis in an opposite direction to disengage the protrusions **237** from the detent areas **225** and pulling the surface treatment head **214** and the first interconnector **230** apart.

[0060] The third detachable connector **240** of the first interconnector **230** carries a female portion of the bayonet connection comprising a cavity **241** with a pair of slots **243** (only one slot **243** is visible in FIG. **11**). The fourth detachable connector **244** of the end interconnector **232** carries a male portion of the bayonet connection comprising a barrel **247** and a pair of protrusions **249** formed on the barrel **247**. As described with respect to the corresponding structures of the first and second detachable connectors **222** and **236**, the cavity **241** and slots **243** of the third detachable connector **240** are configured to receive the barrel **247** and protrusions **249**, respectively, of the fourth detachable connector **244**. The protrusions **249** are received and retained in detent areas **245** provided at the end of each slot **243** (only one detent area **245** is visible in FIG. **11**) to align the first interconnector **230** with respect to the end interconnector **232** and provide a secure, twist-on and twist-off mechanical connection of the first interconnector **230** to the end interconnector **232**.

[0061] The detachable connectors **236B** and **236C** at the first ends **234B** and **234C** of the second interconnectors **230B** and **230C** each carry a male portion of the bayonet connection comprising a respective barrel **235B** and **235C** and pair of protrusions **237B** and **237C** formed on the barrel **235B** and **235C**. The detachable connectors **240B** and **240C** at the second ends **238B** and **238C** of the second interconnectors **230B** and **230C** each carry a female portion of the bayonet connection comprising a respective cavity **241B** and **241C** comprising respective slots **243B** and **243C** with detent areas (not visible). The detachable connectors **236B**, **236C**, **240B**, and **240C** may be removably attached, i.e., via the bayonet connection, as described herein to corresponding ones of

the detachable connectors **222**, **236**, **240**, and/or **244** of the surface treatment head **214**, the first interconnector **230**, and/or the end interconnector **232**.

[0062] The modular personal care implement **200** may be part of a kit (not shown) comprising multiple components, including one or more surface treatment heads **214**, at least one first interconnector **230**, at least one second interconnector **230B** and/or **230C**, and at least one end interconnector **232**. The kit comprising the modular personal care implement **200** may be substantially similar to the kit **80** described with respect to FIG. **12**. As described with respect to the modular personal care implement **10** and **10'**, the modular personal care implement **200** provides an ergonomic handle section **212** that allows a user to customize the configuration as desired, e.g., to adjust a number and/or configuration of the components to obtain a desired length and/or width (not labeled) of the handle section **212** and/or to adjust the number and/or configuration of the components based on hardness, coefficient of friction, surface texture, color, and/or mass to obtain a desired look and feel for the handle section **212**. The surface treatment head(s) **214**, the first interconnector(s) **230**, the second interconnector(s) **230B** and/or **230C**, and the end interconnector(s) **232** may also be easily replaced, as described herein.

[0063] FIG. **13** is a flow diagram of a method **300** of providing a modular personal care implement, in accordance with the present disclosure. As shown in FIG. **13**, the method **300** includes providing one or more surface treatment heads having a first end with an applicator and a second end with a first detachable connector at **310**; providing a plurality of first interconnectors each having a first end with a second detachable connector and a second end having a third detachable connector at **320**; providing a plurality of second interconnectors each having a first end with a fourth detachable connector and a second end having a fifth detachable connector at **330**; and providing one or more end interconnectors each having a first end with a sixth detachable connector at **340**. The method **300** further includes removably attaching the second detachable connector of one first interconnector of the plurality of first interconnectors to the first detachable connector of one surface treatment head of the one or more surface treatment heads at **350**. The method **300** may further include removably attaching the sixth detachable interconnector of one of the one or more end interconnectors to the third detachable connector of the one first interconnector at **360**, after which the method **300** may conclude.

[0064] The method **300** may optionally include removably attaching the fourth detachable connector of one second interconnector of the plurality of second interconnectors to the third detachable connector of the one first interconnector, and removably attaching the sixth detachable interconnector of one of the one or more end interconnectors to the fifth detachable connector of the one second interconnector. In some examples, the one or more surface treatment heads may each further comprise a release mechanism positioned between the first and second end of the surface treatment head, and the method may further include actuating the release mechanism to eject a portion of the one surface treatment head comprising the applicator.

[0065] The modular personal care implement formed by the method **300** may include any of the modular personal care implements **10**, **10'**, **100**, and **200** described herein, in which the first detachable connector of the one or more surface treatment heads, the second and third detachable connectors of the plurality of first interconnectors, the fourth and fifth detachable connectors of the plurality of second interconnectors, and the sixth detachable connector of the one or more end interconnectors may comprise a mechanical connection, i.e., a bayonet connection or a threaded connection, or a magnetic connection.

[0066] The dimensions and values disclosed herein are not to be understood as being strictly limited to the exact numerical values recited. Instead, unless otherwise specified, each such dimension is intended to mean both the recited value and a functionally equivalent range surrounding that value. For example, a dimension disclosed as “40 mm” is intended to mean “about 40 mm.”

[0067] Every document cited herein, including any cross referenced or related patent or application

and any patent application or patent to which this application claims priority or benefit thereof, is hereby incorporated herein by reference in its entirety unless expressly excluded or otherwise limited. The citation of any document is not an admission that it is prior art with respect to any invention disclosed or claimed herein or that it alone, or in any combination with any other reference or references, teaches, suggests, or discloses any such invention. Further, to the extent that any meaning or definition of a term in this document conflicts with any meaning or definition of the same term in a document incorporated by reference, the meaning or definition assigned to that term in this document shall govern.

[0068] While particular embodiments of the present invention have been illustrated and described, it would be obvious to those skilled in the art that various other changes and modifications can be made without departing from the spirit and scope of the invention. It is therefore intended to cover in the appended claims all such changes and modifications that are within the scope of this invention.

Claims

1. A modular personal care implement comprising: a surface treatment head having a first end with an applicator and a second end with a first detachable connector; and a first interconnector having a first end with a second detachable connector and a second end with a third detachable connector, wherein the second detachable connector is removably attached to the first detachable connector of the surface treatment head.
2. The modular personal care implement of claim 1, wherein the first detachable connector of the surface treatment head and the second detachable connector of the first interconnector comprise a mechanical connection or a magnetic connection.
3. The modular personal care implement of claim 1, further comprising: an end interconnector having a first end with a fourth detachable connector, wherein the fourth detachable connector is removably attached to the third detachable connector of the first interconnector.
4. The modular personal care implement of claim 3, wherein the first detachable connector of the surface treatment head, the second and third detachable connectors of the first interconnector, and the fourth detachable connector of the end interconnector comprise a mechanical connection or a magnetic connection.
5. The modular personal care implement of claim 3, wherein the first interconnector comprises a first cross-sectional shape and at least a portion of the end interconnector comprises a second cross-sectional shape that is different from the first cross-sectional shape.
6. The modular personal care implement of claim 3, wherein the first interconnector comprises a first material and the end interconnector comprises a second material that is different from the first material.
7. The modular personal care implement of claim 3, wherein an outer surface of the first interconnector comprises a plurality of gripping protrusions and an outer surface of the end interconnector is smooth.
8. The modular personal care implement of claim 1, further comprising: at least one second interconnector having a first end with a fourth detachable connector and a second end with a fifth detachable connector, wherein the fourth detachable connector is removably attached to the third detachable connector of the first interconnector; and an end interconnector having a first end with a sixth detachable connector, wherein the sixth detachable connector is removably attached to the fifth detachable connector of the at least one second interconnector.
9. The modular personal care implement of claim 8, wherein the first detachable connector of the surface treatment head, the second and third detachable connectors of the first interconnector, the fourth and fifth detachable connectors of the at least one second interconnector, and the sixth detachable connector of the end interconnector comprise a mechanical connection or a magnetic

connection.

10. The modular personal care implement of claim 8, wherein the first interconnector comprises a first length, a first width, a first material, a first mass, a first hardness, and a first color and the at least one second interconnector comprises a second length, a second width, a second material, a second mass, a second hardness, and a second color, and wherein at least one of the second length, second width, second material, second mass, second hardness, or second color is different from a corresponding one of the first length, first width, first material, first mass, first hardness, or first color.

11. The modular personal care implement of claim 1, wherein the surface treatment head comprises a shaving razor cartridge and the applicator comprises at least one blade.

12. The modular personal care implement of claim 1, wherein the surface treatment head comprises a brush head and the applicator comprises a plurality of bristles.

13. The modular personal care implement of claim 1, wherein the surface treatment head further comprises a release mechanism positioned between the first and second end of the surface treatment head, the release mechanism comprising an actuator button configured to eject a portion of the surface treatment head comprising the applicator.

14. A method of providing a modular personal care implement, the method comprising: providing one or more surface treatment heads each having a first end with an applicator and a second end with a first detachable connector; providing a plurality of first interconnectors each having a first end with a second detachable connector and a second end with a third detachable connector; providing a plurality of second interconnectors each having a first end with a fourth detachable connector and a second end with a fifth detachable connector; providing one or more end interconnectors each having a first end with a sixth detachable connector; and removably attaching the second detachable connector of one first interconnector of the plurality of first interconnectors to the first detachable connector of one surface treatment head of the one or more surface treatment heads.

15. The method of claim 14, further comprising: removably attaching the sixth detachable interconnector of one of the one or more end interconnectors to the third detachable connector of the one first interconnector.

16. The method of claim 14, further comprising: removably attaching the fourth detachable connector of one second interconnector of the plurality of second interconnectors to the third detachable connector of the one first interconnector; and removably attaching the sixth detachable interconnector of one of the one or more end interconnectors to the fifth detachable connector of the one second interconnector.

17. The method of claim 14, wherein the first detachable connector of the one or more surface treatment heads, the second and third detachable connectors of the plurality of first interconnectors, the fourth and fifth detachable connectors of the plurality of second interconnectors, and the sixth detachable connector of the one or more end interconnectors comprise a mechanical connection or a magnetic connection.

18. The method of claim 14, wherein the one or more surface treatment heads each further comprise a release mechanism positioned between the first and second end of the surface treatment head, the method further comprising: actuating the release mechanism to eject a portion of the one surface treatment head comprising the applicator.

19. A kit comprising: a first surface treatment head comprising a shaving razor cartridge, the first surface treatment head having a first end with an applicator and a second end with a first detachable connector, wherein the applicator comprises at least one blade; at least one first interconnector having a first end with a second detachable connector and a second end with a third detachable connector; at least one end interconnector having a first end with a fourth detachable connector; a case comprising a top portion and a bottom portion; and an insert comprising one or more cavities configured to receive the first surface treatment head, first interconnector, and end

interconnector, wherein the bottom portion of the case defines a compartment configured to removably receive the insert within the bottom portion.

20. The kit of claim 19, further comprising: a second surface treatment head comprising a brush head, the second surface treatment head having a first end with an applicator and a second end with a first detachable connector, wherein the applicator comprises a plurality of bristles; and at least one second interconnector having a first end with a fifth detachable connector and a second end with a sixth detachable connector, wherein the insert further comprises one or more cavities configured to receive the second surface treatment head and the at least one second interconnector.
